

2) Theory - HW

1) Encrypt "CHASE YOUR DREAMS" using a

a) Caesar cipher with

$$n = 3$$

plain text = "CHASE YOUR DREAMS"

Cipher text = FKDVH BRXU GUHDPV

→ FKDVH BRXU GUHDPV

b) Shift cipher with key 7

$$n = 7$$

plain text = "CHASE YOUR DREAMS"

Cipher text = JOHZZ FVBY KYLHTZ

→ JOHZZ FVBY KYLHTZ

c) ROT 13

$$n = 13$$

plain text = CHASE YOUR DREAMS

Cipher text = PUNER LBHE QERNZ F

→ PUNER LBHE QERNZ F

2) Decryption

a) "WRWDOOB VHFUHW PHVVOJH" using ceasar cipher.

$n=3$

Cipher text = WRWDOOB VHFUHW PHVVOJH

Plain text = TOTALLY SECRET MESSAGE

→ TOTALLY SECRET MESSAGE

b) "MJQQT BTWQI" using shift cipher, 5

$n=5$

Cipher text = MJQQT BTWQI

Plain text = HELLO WORLD

→ HELLO WORLD

b) "SJB DTWP YNRJ" using shift cipher, 5

$n=5$

Cipher text = SJB DTWP YNRJ

Plain text = NEW YORK TIME

→ NEW YORK TIME

c) "FRAQ ZBER PBSSRR" using ROT13

$n=13$

Cipher text = FRAQ ZBER PBSSRR

Plain text = SEND MORE COFFEE

→ SEND MORE COFFEE